

Hasan Mahmood

Current Position: **PhD Scholar / Co-Manager, Nova Scotia Math Circles Program**
Affiliation: Department of Mathematics and Statistics, Dalhousie University
Halifax, NS, Canada
Inactive Position: **Associate Professor of Mathematics**
Affiliation: Department of Mathematics, Government College University Lahore (Pakistan)
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Education **Department of Mathematics and Statistics, Dalhousie University**
Halifax, Nova Scotia (Canada)
Ph.D. Scholar. (2023 - present)

Abdus Salam School of Mathematical Sciences, Government College
University Lahore (Pakistan)
Ph.D., 2012. (Thesis Title: Smooth/Impulsive Linear Systems)

Department of Mathematics, Government College University Lahore
M.Sc. Mathematics, 2006 (completed with Gold Medal and Academic Roll of)
Honour with CGPA of 3.64 out of 4.0).

University of the Punjab
Bachelor of Arts (Liberal), 2003.

Research Interests Combinatorics
Commutative Algebra
Algebraic Geometry
Algebraic Combinatorics
Systems Theory
Mathematics Education.

Professional Experience

04/2026 – Present	Co - Manager & Research Assistant (RA) Nova Scotia Math Circles Program, Dalhousie University, NS, Canada
09/2026 – 12/2026	RA & University Mathematics Instructor (PT) Nova Scotia Math Circles Program, Dalhousie University, NS, Canada
05/2026 – 06/2026	RA & University Mathematics Instructor (PT) Nova Scotia Math Circles Program, Dalhousie University, NS, Canada
01/2026 – 4/2026	Teaching Assistant (TA) & RA & Math Circles Presenter Dalhousie University, Halifax, NS, Canada
09/2025 – 12/2025	TA & RA & Math Circles Presenter Dalhousie University, Halifax, NS, Canada
07/2025 – 08/2025	RA & University Mathematics Instructor (PT) Dalhousie University, Halifax, NS, Canada
01/2025 – 04/2025	TA & RA & University Mathematics Instructor (PT) Math Circles Presenter Dalhousie University, Halifax, NS, Canada
09/2024 – 12/2024	RA & University Mathematics Instructor (PT) Dalhousie University, Halifax, NS, Canada
09/2024 – 12/2024	RA & University Mathematics Instructor (PT) Mount Saint Vincent University, Halifax, NS, Canada
05/2024 – 08/2024	RA & University Mathematics Instructor (PT) Dalhousie University, Halifax, NS, Canada
01/2024 – 04/2024	TA & RA & University Mathematics Instructor (PT) Math Circles Presenter Dalhousie University, Halifax, NS, Canada
09/2023 – 12/2023	TA & RA & Math Circles Presenter Dalhousie University, Halifax, NS, Canada
02/2021 – 08/2023	Associate Professor (on unpaid leave now) Department of Mathematics, GC University Lahore, Pakistan
08/2019 – 08/2020	Visiting Postdoc / Adjunct Assistant Professor Department of Mathematics, Tulane University, New Orleans, LA, USA
06/2014 – 02/2021	Assistant Professor Department of Mathematics, GC University Lahore, Pakistan
08/2007 – 06/2014	Lecturer Department of Mathematics, GC University Lahore, Pakistan

Research / Journal Publications (selected)

- [1] Cooper, S. M., Faridi, S., Mahmood, H., 2026. Simplicial resolutions of the quadratic power of monomial ideals. *Communications in Algebra*, 1–16.
<https://doi.org/10.1080/00927872.2026.2715474>
- [2] Bu, L., Faridi, S., Hewalage, M., Holleben, T., **Mahmood, H.**, Wesley, S., Veer, D., Wang, K., 2025. When are Morse Resolutions Polyhedral? (submitted).
- [3] Bilal, H.M, Ahmed, S., **Mahmood, H.**, Binyamin, M.A., 2025. Expected $\mathcal{NF}$ -number of Disjoint Union of Finite Copies of Complete Graph, *Ricerche di matematica*.
<https://doi.org/10.1007/s11587-025-00987-5>
- [4] Binyamin, M. A., Adeel, Alali, A. S., **Mahmood, H.**, 2025. On Connectedness, Girth and Diameter of Gap Poset Graphs, *Proceedings of the Bulgarian Academy of Sciences*, 78 (6), 813-820.
- [5] Hanif, M. F., **Mahmood, H.**, Ahmed, S., 2024. On degree-based entropy measure for zero-divisor graphs. *Discrete Mathematics, Algorithms and Applications*.
(<https://doi.org/10.1142/S1793830923501045>).
- [6] Rehman, F. U., **Mahmood, H.**, Nguyen, T., Binyamin M. A., 2023. Generalization of f -Graphs and Their Algebraic Aspects, *Journal of Mathematics*, *Journal of Mathematics Volume 2023*, Article ID 7984489, 9 pages <https://doi.org/10.1155/2023/7984489>.
- [7] Hibi, T., **Mahmood, H.**, 2022. The $\mathcal{NF}$ -Number of a Simplicial Complex, *Algebra Colloquium*, 29 (4), 643-650.
- [8] Hanif, M. F., **Mahmood, H.**, Manzoor, S., Petros, F. B., 2022. Computing the Entropy Measures for the Line Graphs of Some Chemical Networks, *Computational Intelligence and Neuroscience*, Volume 2022, Article ID 2006574, 18 pages.
- [9] Huy, T.H., Keimper, G., **Mahmood, H.**, O'Rourke, J. L., 2022. Density of f -Ideals and f -Ideals in Mixed Small Degree, *Mathematica Scandinavica* 128 (1), 14-34.
- [10] Bilal, H.M, Ahmed, S., **Mahmood, H.**, Binyamin, M.A., 2022. The $\mathcal{NF}$ -Number of Two Complete Graphs Joined by a Common Vertex, *Comptes rendus de l'Academie bulgare des Sciences*, 75 (5), 640-648.
- [11] Binyamin, M.A, Ali, W., Aslam, A., **Mahmood, H.**, 2022. A Complete Classification of Planar Graphs Associated with the Ideal of the Numerical Semigroup, *Iranian Journal of Science and Technology*, *Transactions A: Science*, 46, pages 491–498.
- [12] Bilal, H.M, Ahmed, S., **Mahmood, H.**, Binyamin, M.A., 2022. The $\mathcal{NF}$ -Number of n Disjoint Copies of Complete Graphs (submitted).
- [13] Rehman, F.U., Hasan, H., **Mahmood, H.**, Binyamin, M.A., 2021. Characterization and Newton Complementary Dual of Quasi f -Ideals (submitted) (arxiv: 2101.01584v1).

- [14] Aslam, S., Binyamin, M.A., **Mahmood, H.**, 2021. A Classifier For Equidimensional Contact Unimodal Map Germs. *Bulletin Mathematique de la Societe des Sciences Mathematiques de Roumanie*, Volume 64 (112), Issue no. 3, 285-300.
- [15] Liu, Y., Binyamin, M.A., Aslam, S., Arshad, M., Fan, C., **Mahmood, H.**, Liu, J-B., 2021. An Implementation of Lipschitz Simple Functions in Computer Algebra System Singular. *Complexity*, Volume 2021, Article ID 6695461, 5 pages.
- [16] **Mahmood, H.**, Rehman, F.U., Nguyen, T., Binyamin, M.A., 2020. Quasi f -Simplicial Complexes and Quasi f -Graphs (arxiv:2009.03641v1).
- [17] Xu, P., Binyamin, M.A., Aslam, A., Ali, W., **Mahmood, H.**, Zhou, H., 2020. Characterization of Graphs Associated with the Ideal of Numerical Semigroups. *Journal of Mathematics*, vol. 2020, Article ID 6094372 (6 pages).
- [18] Binyamin, M.A., Aslam, S., Mehmood, K., **Mahmood, H.**, Ishaq, M., 2020. Recognition of Contact Map Germs from the Plane to the Plane by Invariants, *Comptes rendus de l'Academie bulgare des Sciences*, 73(6) 758-766.
- [19] **Mahmood, H.**, Rehman, F.U., Binyamin, M.A., 2020. A Note on f -Graphs, *Journal of Algebra and Its Applications*. 19-10, p.2050193 (9 pages).
- [20] Shahzad, A., Binyamin, M.A., **Mahmood, H.**, 2020. Characterization of Hypersurface Singularities in Positive Characteristic, *Revista de la Unión Matemática Argentina*, 61-2, 457-468.
- [21] Mehmood, T., **Mahmood, H.**, Hussain, M., 2018. Edge Irregularity Strength of Some Rooted Product Graphs, *Utilitas Mathematica*, Volume 107, 279-286.
- [22] Binyamin, M.A., **Mahmood, H.**, Kanwal, S., 2017. On the Classification of Simple Maps from Plane to Plane, *Journal of Algebra and Its Applications*, 16-10, p.1750199 (8 pages).
- [23] Binyamin, M.A., Rabia, **Mahmood, H.**, Khan, J.A., Mehmood, K., 2017. Characterization of Uni-modal Parametric Plane Curve Singularities, *Journal of Algebra and Its Applications*, 16-2, 1750039 (10 pages).
- [24] Mehmood, T., **Mahmood, H.**, Hussain, M., 2017. Edge Irregularity Strength of Multi Middle and Extended Prism Graphs, *Journal of Computational and Theoretical Nanoscience*, Vol. 14, 5248-5252.
- [25] **Mahmood, H.**, Anwar, I., Binyamin, M.A. and Y, S., 2017. On the connectedness of f -simplicial complexes. *Journal of Algebra and Its Applications*, 16-1 p.1750017-25.
- [26] **Mahmood, H.**, 2016. A Note on f -Ideals. Conference Paper, presented at Canadian Mathematical Society's 2016 Winter Meeting, Dec 2-5, Niagara Falls, Canada.
- [27] Binyamin, M.A., Khan, J.A., **Mahmood, H.**, 2015. A Classifier for Ideal Unimodular Singularities, *Analele Stiintifice ale Universitatii Ovidius Constanta, Seria Matematica*, 23-2, 59-69.
- [28] Anwar, I., **Mahmood, H.**, Binyamin, M.A. and Zafar, M.K., 2014. On the Characterization of f -Ideals. *Communications in Algebra*, 42(9), pp.3736-3741.

- [29] **Mahmood, H.**, Anwar, I. and Zafar, M.K., 2014. A construction of Cohen-Macaulay f-graphs. *Journal of Algebra and Its Applications*, 13(06), p.1450012 (8 pages).
- [30] Akram, M.S., Lomadze, V., **Mahmood, H.**, Zafar, M.K., 2014. addendum to '(Singular) states models and (singular) LTID systems', *International Journal of Control*, 87-6, 1312-1315.
- [31] Akram, M.S., Lomadze, V., **Mahmood, H.**, Zafar, M.K., 2013. (Singular) states models and (singular) LTID systems, *International Journal of Control*, 87-3, 567-580.
- [32] Akram, M.S., Lomadze, V., **Mahmood, H.**, Zafar, M.K., 2012. Smooth/Impulsive Linear Systems: Interconnections. *International Journal of Control*, 86-3, 416-425.
- [33] Lomadze, V., **Mahmood, H.**, 2011. Smooth/Impulsive Linear Systems: Controllability. *International Journal of Control*, 84, 679-692.
- [34] Lomadze, V., **Mahmood, H.**, 2010. Smooth/Impulsive Linear Systems: Axiomatic description. *Linear Algebra and its Applications*, 433, 1997-2009.

Teaching Undergraduate Level Courses

BS (Hons) Courses:

- **MATH 3031: Abstract Algebra - I**, Fall 2026 Department of Mathematics and Statistics, Dalhousie University (currently teaching).
- **MATH 1030: Linear Algebra and Matrix Theory I** — (Online), Summer 2026 Department of Mathematics and Statistics, Dalhousie University.
- **MATH 1290: Engineering Mathematics II** — (Online), Summer 2025, Department of Mathematics and Statistics, Dalhousie University.
- **CSCI/MATH 2113: Discrete Structures II** — Winter 2025, Department of Mathematics and Statistics, Dalhousie University.
- **MATH 1102: Precalculus I** — Fall 2024, Department of Mathematics and Statistics, Mount Saint Vincent University.
- **MATH 3300: Optimization** — Fall 2024, Department of Mathematics and Statistics, Dalhousie University.
- **MATH 1030: Linear Algebra and Matrix Theory I** — (Online), Department of Mathematics and Statistics, Dalhousie University.
- **CSCI 2360: Probability and Statistics for Computer Science** — Winter 2024, Faculty of Computer Science, Dalhousie University.
- **Long Calculus II** — Summer 2020 (Online), Tulane University.
- **Abstract Algebra II** — Spring 2020 (Hybrid), Tulane University.

Courses Taught at Government College University (GCU), Lahore, Pakistan:

- **Precalculus** — Spring 2013.
- **Calculus and Analytic Geometry** — Winter 2014.
- **Calculus I** — Winter 2021.
- **Multivariate Calculus** — Winter 2018, 2017, 2015.
- **Discrete Mathematics** — Winter 2016, 2014, 2013.
- **Number Theory** — Spring 2022, 2018, 2017, 2015.
- **Algebra I** — Spring 2018, 2017, 2014.
- **Algebra II** — Winter 2018, 2017, 2015, 2014.
- **Linear Algebra** — Spring 2018, 2015, 2014.
- **Rings and Modules** — Winter 2021, 2018, 2017, 2015, 2014.
- **Real Analysis I** — Winter 2016.
- **Vector and Tensor Analysis** — Winter 2014.
- **Complex Variables and Transforms** — Winter 2015.
- **Topology** — Spring 2019.
- **Graph Theory** — Spring 2019.
- **Elementary Mathematics** — Spring 2013–2019 (multiple terms).
- **Calculus III** — Taught multiple terms.
- **Mathematical Statistics** — Winter 2020, Winter 2021; taught multiple additional terms.
- **Precalculus** — Spring 2013.
- **Calculus and Analytic Geometry** — Winter 2014.
- **Calculus I** — Winter 2021.
- **Multivariate Calculus** — Winter 2018, Winter 2017, Winter 2015.
- **Discrete Mathematics** — Winter 2016, Winter 2014, Winter 2013.
- **Number Theory** — Spring 2022, Spring 2018, Spring 2017, Spring 2015.
- **Algebra I** — Spring 2018, Spring 2017, Spring 2014.
- **Algebra II** — Winter 2018, Winter 2017, Winter 2015, Winter 2014.
- **Linear Algebra** — Spring 2018, Spring 2015, Spring 2014.

- **Rings and Modules** — Winter 2021, Winter 2018, Winter 2017, Winter 2015, Winter 2014.
- **Real Analysis I** — Winter 2016.
- **Vector and Tensor Analysis** — Winter 2014.
- **Complex Variables and Transforms** — Winter 2015.
- **Topology** — Spring 2019.
- **Graph Theory** — Spring 2019.
- **Elementary Mathematics** — Spring 2013–2019 (multiple terms).

College Level Math Courses taught at GC University:

- **Algebra and Trigonometry** (Grade-11 one year course); taught two times, 2012-2013, 2013-2014.
- **Calculus and Analytical Geometry** (grade-12 one year course); taught four times, 2012-2013, 2013-2014, 2014-2015, 2015-2016.

Mentorship: PhD Thesis Supervision

- Tariq Mehmood
Thesis Title: Irregular Labeling of Some Families of Graphs (May 2019)
- Fazal Ur Rehman
Thesis Title: F-Ideals and Quasi F-Ideals (June 2024)

Master's Thesis Supervision

- Benish Arshad
Thesis Title: New Characterizations of f -Ideals of Degree 2 (January 2014)
- Muhammad Nazim
Thesis Title: Spanning Simplicial Complexes of General Friendship Graphs (January 2014)
- Saba Yasmeen
Thesis Title: f -Simplicial Complexes (August 2015)
- Maiza Rouf
Thesis Title: Unmixed Squarefree Monomial Ideals of Certain Heights (February 2016)
- Mehak Sohail
Thesis Title: Matroidal Ideals of Some Graphical Matroids (February 2016)

- Sadaf Zahoor
Thesis Title: A Survey on Smooth/Impulsive Linear Systems (July 2016)
- Aziz Ur Rehman
Thesis Title: Minimal Vertex Covers of Some Families of Graphs (February 2017)
- Seema Mazhar
Thesis Title: Properties of f -Ideals: A Survey (July 2017)
- Syeda Mehnaz Fatima
Thesis Title: Some Properties of f -Simplicial Complexes (January 2018)
- Sehar Gul Rani
Thesis Title: Cover Complexes of Some Families of Graphs (January 2018)
- Ayesha Liaqat
Thesis Title: Simplicial Complexes on a Chessboard (January 2019)
- Kulsoom Mehak
Thesis Title: Cover Complexes of Star, Friendship, and Book Graphs (January 2019)
- Saba Saif
Thesis Title: Independent Complexes of Some Families of Graphs (January 2019)
- Ahsan Ali
Thesis Title: Independent Complexes of f -Graphs (April 2021)
- Usman Ghanni
Thesis Title: Chessboard Graphs (June 2021)
- Tajamal Mazhar
Thesis Title: A Study of Quasi f -Ideals and Their Properties (May 2022)
- Tamasam Hadayat
Thesis Title: Construction of Non-Homogeneous Quasi f -Ideals of Type (a, b, c) (May 2022)
- Hamna Younas
Thesis Title: On the $\mathcal{NF}$ -Number of a Complete Graph with a Whisker (November 2022)
- Sidra Choudhry
Thesis Title: Gröbner Bases and Primary Decomposition of Binomial Edge Ideals of Some Graphs (November 2022)
- Hina Komal
Thesis Title: Edge-Neighborhood Complexes of Some Families of Graphs (November 2022)
- Abdul Rehman
Thesis Title: New Codes and Programming in Macaulay2 (November 2022)

Administrative Services

- Serving as Teaching Seminar Co-organizer, Department of Mathematics and Statistics, Dalhousie University (Fall 2025–present). Co-organizing the Teaching Seminar Series with Dr. Sarah Chisholm.
- Co-Manager of the Nova Scotia Math Circles program. Organize and Participate in outreach initiatives by visiting schools across the province and delivering engaging mathematics presentations to students in Grades 1–12.
- I have served as the GCU's departmental focal person for the United Nations' Sustainable Development Goals Data Management (2020-2023)
- I have served as the departmental focal person for the GCU's International Ranking Data.
- President, GC University Chess Club. I am also a Chess Player. My rating at Chess.com is around 1900.
- Member Of the Sports Board of GC University.
- I was also a member of the Departmental Board of Studies, Department of Mathematics, GC University Lahore.
- I served as Time Table Incharge for the Department of Mathematics, GC University Lahore, Pakistan, from February 2016 to July 2019.
- I was the Co-ordinator of M.Sc. Evening Program of the Department of Mathematics, GC University Lahore, Pakistan, from September 2013 to August 2018.
- I remained the Departmental Seminar Incharge of the Department of Mathematics, GC University Lahore, Pakistan, for the period: May 2012 - Nov 2017.
- I served as the Co-Advisor of GCU's Chawala Mathematical Society for one year, from May 2018 to July 2019.
- Secretary, Academic Counsel of Mathematics Department, Government College University Lahore (May 2018 to July 2019).
- Member of the Departmental Sports Committee (September 2017 to July 2019).
- Member of the Departmental Budget Committee (May 2018 to July 2019).
- Member of the Admission Committee of the Mathematics Department.

Honors and Awards

- I served as the President of the Chess Club at GC University Lahore, Pakistan, for six years (2013–2019).

- I was one of the Olympiad Referees for “5th CASPAM Regional Student Olympiad of Mathematics” organized by BZ University, Multan, on April 19, 2018.
- I was one of the Olympiad Referees for “5th and 6th CASPAM Regional Student Olympiad of Mathematics” organized by BZ University, Multan, on April 16, 2019.
- I was awarded a prize of Rs. 100,000/- from the National Mathematical Society in its Award Distribution Ceremony in March 2016. It was as an appreciation for being the first Pakistani who brought got a medal from the international undergraduate math contest.
- I won the First Prize of Best Young Speaker on presenting my contributed paper in the “5th World Conference on 21st Century Mathematics 2011” at Abdus Salam School of Mathematical Sciences, Government College University Lahore, Pakistan.
- I received a Gold Medal and Academic Roll of Honor in M.Sc. (Mathematics) from GC University Lahore. (2006)
- I Won a Bronze Medal in the “11th International Scientific Olympiad on Mathematics 2006” held in Tehran, The Islamic Republic of Iran.
- I was Selected as a member of Pakistan’s five membered national team for international mathematics competition after standing at 2nd position amongst the contestants of 17 universities of Pakistan. (2006)

My Talks

- “Some Fun Counting Through Recursion”, Dalhousie Math Camp, Dalhousie University, July 23, 2026.
- Presented a talk titled “ h -Vectors and Algebraic Properties of Stanley–Reisner Rings” at the Commutative Algebra Seminar Series, Dalhousie University, February 28, 2026.
- Presented a talk titled “Face Rings of Simplicial Complexes and Their Hilbert Functions” at the Commutative Algebra Seminar Series, Dalhousie University, February 14, 2026.
- Presented a talk titled “Simplicial Resolutions of Powers of Monomial Ideals” at the Canadian Mathematical Society (CMS) Winter Meeting 2025, Toronto, December 5–8, 2025.
- Presented a talk titled “The Long-Term Behavior of Associated Primes in Powers: Stability and Persistence” at the Commutative Algebra Session, Department of Mathematics and Statistics, Dalhousie University, March 4, 2025.
- Presented a talk titled “Comparing f -Vectors: Facet Complex vs. Non-Face Simplicial Complex” at the Commutative Algebra Session, Department of Mathematics and Statistics, Dalhousie University, January 28, 2025.
- Presented a talk titled “Stability of Associated Primes of Powers of an Ideal” at the Commutative Algebra Seminar Series, Dalhousie University, January 14, 2025.

- Presented a talk titled “Mutations of Simplicial Complexes” at the Canadian Mathematical Society Winter Meeting 2023, Montreal, December 3, 2023.
- Presented a talk titled “F-Ideals, F-Graphs, and F-Simplicial Complexes” in the Graduate Seminar Series, Dalhousie University, November 3, 2023.
- Presented a talk titled “F-Ideals and Quasi F-Ideals” in the Algebra Seminar Series, McMaster University, December 10, 2019.
- Presented a talk titled “F-Ideals, F-Graphs, and F-Simplicial Complexes” in the Algebra and Combinatorics Seminar Series, Tulane University, October 2, 2019.
- Presented a talk titled “A Note on F-Ideals” at the Canadian Mathematical Society Winter Meeting 2016, Niagara Falls.
- Delivered an invited talk on “Monomial Ideals” at Government Degree College for Women, Lahore, Pakistan, April 6, 2019.
- Presented a talk titled “ f -Ideals and f -Simplicial Complexes” at the Garrison International Conference on Pure and Applied Mathematics, April 1–3, 2019.
- Presented a talk at the One-Day Symposium on Algebra and Number Theory, organized by COMSATS University Islamabad (Lahore Campus), October 13, 2018. Professor Francesco Pappalardi (ICTP) was an invited speaker at the symposium.
- Delivered an invited lecture on “Graphs and Ideals” at Government College (Satellite Town), Gujranwala, Pakistan, June 12, 2018.
- Presented a seminar titled “Primary Decomposition and Some of Its Applications in Combinatorics” at the Department of Mathematics, Government College University Lahore, December 19, 2017.
- Delivered an invited talk at the One-Day Symposium on Combinatorial Commutative Algebra, CIIT Islamabad, July 20, 2017.
- Presented a contributed paper at the “5th World Conference on 21st Century Mathematics 2011,” Abdus Salam School of Mathematical Sciences, Government College University Lahore, Pakistan.

Participation

- Participated in the SLMath Graduate Summer School on Singularities in Commutative Algebra Through Cohomological Methods, held at the Simons Laufer Mathematical Sciences Institute (SLMath), Berkeley, California, USA (July 6–17, 2026).
- Participated in the Séminaire de Mathématiques Supérieures (SMS) 2025: An Introduction to Recent Trends in Commutative Algebra, held at the Fields Institute, University of Toronto (June 2–13, 2025).

- Attended the Workshop on the Applications of Commutative Algebra, held at the Fields Institute, University of Toronto, from May 26–30, 2025.
- Attended the Graph Searching in Canada (GRASCan) Workshop 2024, held on August 8–9, 2024.
- Attended Combinatorial Algebra meets Algebraic Combinatorics (CAAC) hosted by LACIM in collaboration with the CRM-ISM University of Quebec Montreal, January 26-28, 2024.
- Attended Canadian Mathematical Society's 2023 Winter Meeting held in Montreal during Dec 2-4, 2023.
- Attending ICTP's (Italy) online Graduate Summer Course on Tight Closure of Ideal and Its Applications (May 16 - July 25, 2022).
- Attending CIMPA Research School on Algebraic and Combinatorial Methods in Geometry, March 1-11, 2022, hosted by LUMS, Lahore.
- Attending one day workshop on "A Successful Journey From Research to Publication" held on February 17, 2022, organized by ORIC, GCU Lahore.
- successfully completed 5 Weeks Online Teaching Training Program organized by the Center for Engaged Learning and Teaching (CELT) and the Innovative Learning Center (ILC), Tulane University, May 22 - June 23, 2020, New Orleans, USA.
- Attended Southwest Local Algebra Meeting Organized by Tulane University, March 7-8, 2020, New Orleans, USA.
- Attended Canadian Mathematical Society Winter Meeting, Dec 6-9, 2019, Toronto, Canada.
- Attended 2019 Workshop on Solitons and Its Applications, organized by the Department of Mathematics, GC University Lahore, on January 07, 2019.
- Attended the International Conference on Mathematics and its Applications organized by the Department of Mathematics, GCU Lahore, November 13-15, 2018.
- Attended One Day Workshop on Non-Linear Analysis organized by GCU Lahore on July 17, 2018.
- Attended the '2018-Workshop on Fluid Dynamics' organized by the Department of Mathematics, GC University Lahore, on 15-01-2018.
- Attended the International Conference on Mathematics and its Applications organized by the Department of Mathematics, GCU Lahore, November 13-15, 2017.
- Attended One Day Symposium on Combinatorial Commutative Algebra organized by CIIT Islamabad on 20-07-2017.
- Participated and presented a talk A NOTE ON F-IDEALS at CMS Winter Meeting, Dec 2-5 2016, Niagara Falls.

- Participated in the ‘ 16th International Pure Mathematics Conference’ held on August 21-24, 2015 at Margala Hotel Islamabad.
- Participated in Workshop on “Recent Advances in Commutative Algebra and Algebraic Geometry” held on March 28-29, 2014 at Centre for Advanced Studies on Mathematics (CASM) LUMS Pakistan.
- Participated in the “6th World Conference on 21st Century Mathematics; March 6-9, 2013” ASSMS GC University Lahore.
- Participated in the “16th Master Trainer Faculty Professional Development Programme (MT-FPDP)” held on May 23-25, 2012 at GC University Lahore.
- Successful Participation in the “CIMPA-ICTP-UNESCO- MESR-MICINN-PAKISTAN Research School on Local Analytic Geometry; Feb 4-13, 2012” ASSMS GC University Lahore .
- Participated in Conference on “Recent Advances in Mathematical Methods, Models and Applications” held on April 16-17, 2011 at Centre for advanced studies on Mathematics (CASM) LUMS Pakistan.
- Participated in “One Day Symposium on Mathematics and its Applications “held on April 02, 2011 at COMSATS Institute of Information Technology Lahore.
- Participated in the “5th World Conference on 21st Century Mathematics; Feb 9-13, 2011” ASSMS GC University Lahore.
- Active Participation in the “ASSMS-NCM-CIMPA-HEC Workshop on Computational Aspects of Singularity Theory; July 22-26, 2010” ASSMS GC University Lahore.
- Valuable participation in the in “4th World Conference on 21st Century Mathematics March 4-8, 2009” ASSMS GC University Lahore.
- Successful Participation in the “CIMPA-UNESCO-IMU-ICTP School on Combinatorial and Computational Aspects of Commutative Algebra; Feb 21-28, 2009” ASSMS GC University Lahore.
- Participation in the “Himalaya School Conference on Geometric, Homological and Combinatorial Aspects in Commutative Algebra; August 20-30, 2008.” ASSMS GC University Lahore.